Name	Prompt
`3-question`	`Write 3 questions that test understanding of different parts of the context. Answer with just the 3 questions and options (do not say the correct answer), each one separated with 2 newlines.`
`summarize`	`Summarize the main points of the context.`
`structure_json`	`Structure the information in JSON form and include all important details like dates, times, names, and numerical values.`
`aggregate`	`Aggregate all the key facts mentioned in the context.`

Table 6. **Prompt templates used in self-study.**

Prompts. We use the prompt templates in Table 6.

We obtain one conversation starter directly from each of `summarize`, `structure_json`, and `aggregate`. For `3-question`, we sample a response from instance A and split the generated output into three conversations starters. We then sample one response per starter.

D. Impact of Reference Queries

We present uniform compaction results under different reference query generation strategies in Figure 5. We evaluate log(Perplexity) of the original cache's answers on 20 QuALITY instances (357 questions). We plot this rather than downstream accuracy because it has lower variance and correlates well with downstream accuracy.

For random vectors, we sample IID from a normal distribution with mean 0 and standard deviation 1, and then apply q-norm scaling for the respective attention heads. We omit this step in `random-vectors-no-qnorm`. In `ss-5k` and `ss-plus-repeat-5k`, we subsample $5{,}000$ query vectors uniformly at random after running the corresponding query generation method.

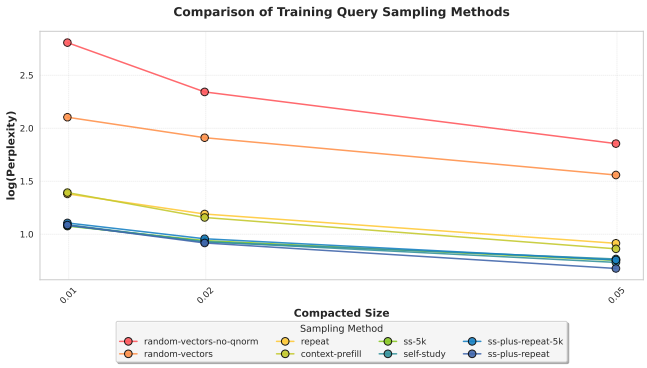

Figure 5. **Reference query sampling comparison.** We compare eight variants for sampling queries. Self-study-based methods perform best, especially at the greatest compaction ratios, with repeat and context-prefill close behind. Subsampling reference queries preserves performance.

E. Head Budgets

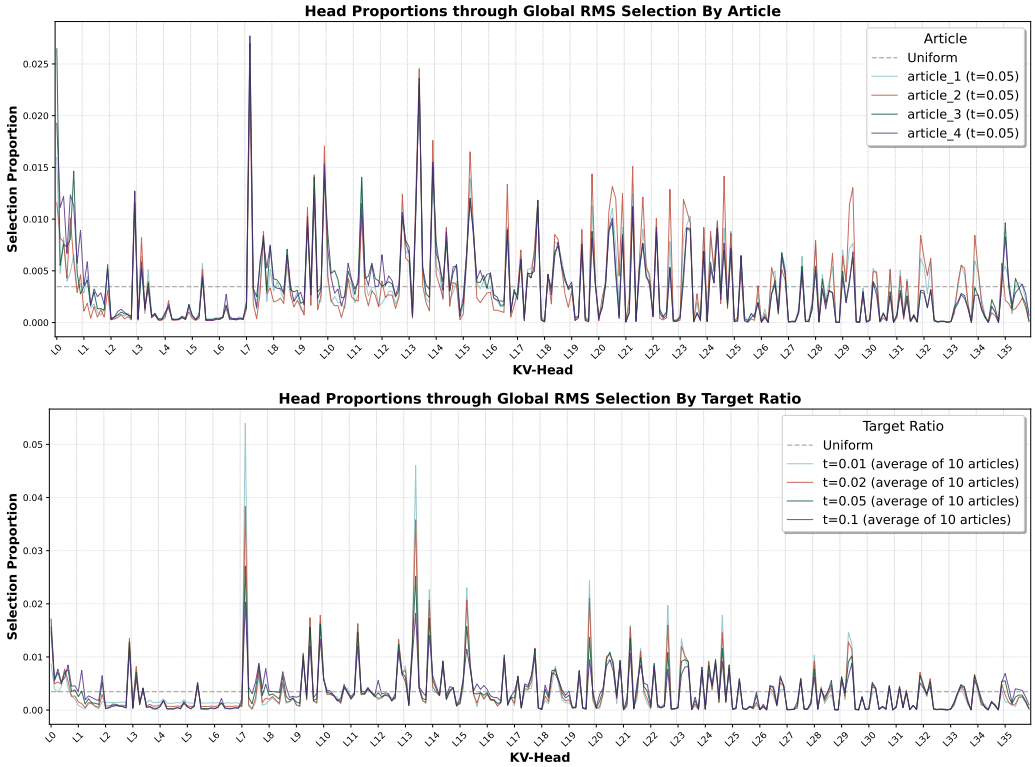

Figure 6. **Head proportions induced by global highest-attention selection.** We plot the fraction of total selected keys attributed to each of the 288 KV heads (36 layers, 8 heads per layer) in Qwen3-4B. **Top:** proportions across 10 QuALITY articles at a fixed global compaction ratio of 0.05. **Bottom:** average proportions across multiple global compaction ratios. The low variance across articles and target ratios indicates stable head importance patterns.

This appendix provides additional evidence motivating nonuniform KV compaction and visualizes the per-head budgets learned by our method.

Global key selection induces stable head proportions. KVzip (Kim et al., 2025b) selects keys based on highest attention scores *globally* across all KV heads, without enforcing per-head quotas. Under a fixed global top-k budget, this induces an implicit allocation in which different heads retain different fractions of the total KV capacity.

Figure 6 shows the implicit head proportions induced by global highest-attention selection in Qwen3-4B, supporting the use of a fixed, precomputed nonuniform schedule.

Head sensitivity curves. Following Section 3.4, we directly measure head sensitivity by varying the compaction ratio of individual heads while holding all other heads fixed. Figure 2 in the main paper shows representative sensitivity curves for Qwen3-4B.

To demonstrate that this behavior generalizes across model families, Figure 7 shows analogous head sensitivity curves for Llama-3.1-8B-Instruct. In both models, we observe substantial differences across heads: some heads are largely insensitive to KV capacity, while others benefit significantly from retaining additional keys.

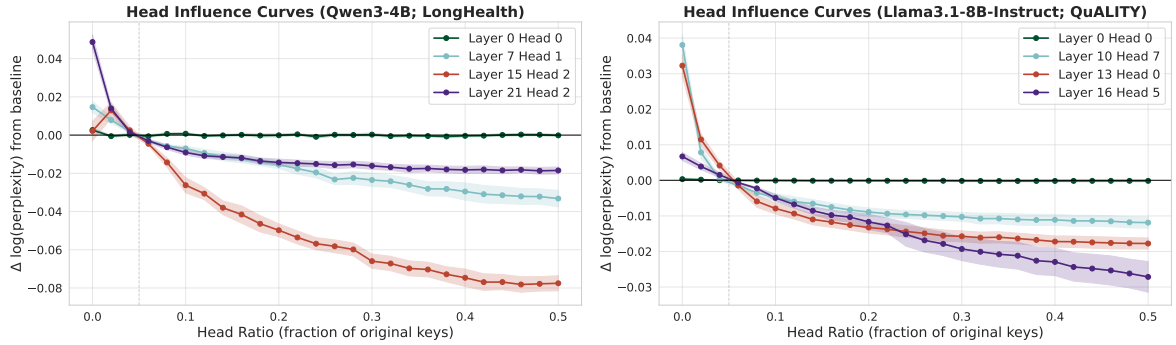

Figure 7. **More Head sensitivity curves.** Analogous to Figure 2, showing loss change as a function of per-head compaction ratio for four selected heads. **Left:** Qwen3-4B with LongHealth exhibits similar patterns to Figure 2. **Right:** Llama-3.1-8B-Instruct also shows head-dependent sensitivity patterns: some heads are largely insensitive to KV capacity, while others benefit substantially from retaining more keys.

Optimized head budgets. Using these sensitivity curves, we apply the budget-allocation procedure described in Section 3.4 and Algorithm 4 to derive model-specific nonuniform head budgets. Figure 8 compares the resulting optimized head proportions to the average head proportions induced by global highest-attention key selection.

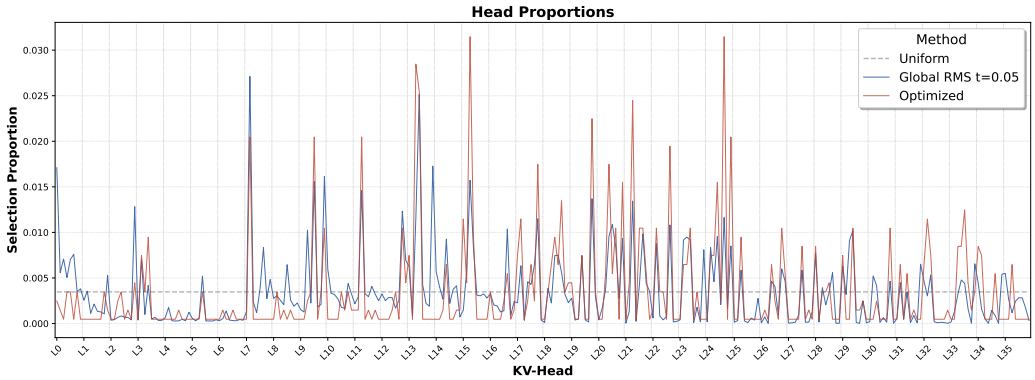

Figure 8. **Optimized head budgets.** We compare the head proportions learned by our nonuniform allocation algorithm to those induced by global highest-attention key selection. The optimized schedule has similar structure as the global highest attention scores schedule. The optimized schedule tends to assign more budget to later layers and less to earlier layers, contrary to PyramidKV (Cai et al., 2025).

Algorithm 4 Nonuniform KV Allocation via Greedy Swaps

Require: Heads $h = 1, \ldots, H$; step size $\eta > 0$ (share units); per-head sensitivity curves $J_h(\rho)$ mapping compaction ratio $\rho \in [0, 1]$ to loss; reference overall compaction ratio r_0 (we use $r_0 = 0.05$).
Ensure: Shares $\{p_h\}$ with $\sum_h p_h = 1$ and $p_h \geq 0$.

Initialize. $p_h \leftarrow \frac{1}{H}$ for all h.
repeat
  Map shares to ratios. $\rho_h \leftarrow p_h \cdot H \cdot r_0$ for all h.
  Compute marginals. For each head h,

$$g_h^+ \leftarrow \begin{cases} J_h(\rho_h) - J_h(\rho_h + \eta'), & \text{if } \rho_h + \eta' \leq 1, \\ -\infty, & \text{otherwise,} \end{cases} \qquad g_h^- \leftarrow \begin{cases} J_h(\rho_h - \eta') - J_h(\rho_h), & \text{if } \rho_h \geq \eta', \\ +\infty, & \text{otherwise,} \end{cases}$$

  where $\eta' = \eta \cdot H \cdot r_0$ is the step size in ratio space.
  Select best swap. $b^\star \leftarrow \arg\max_h g_h^+, \quad a^\star \leftarrow \arg\min_h g_h^- \quad$ with $a^\star \neq b^\star$.
  if $g_{b^\star}^+ > g_{a^\star}^-$ **then**
    $p_{a^\star} \leftarrow p_{a^\star} - \eta; \quad p_{b^\star} \leftarrow p_{b^\star} + \eta.$
  else
    break {No improving swap remains.}
  end if
until termination
return $\{p_h\}$.

F. Further Results

F.1. Mean vs. RMS vs. Max Aggregation

Several prior methods—including H2O, SnapKV, PyramidKV, KVzip, and our *AM-HighestAttnKeys*—select a subset of keys based on attention scores under a collection of query vectors. Concretely, for each key K_j and reference query $\boldsymbol{q}_i$, we compute the attention weight

$$a_{i,j} = \mathrm{softmax}(\boldsymbol{q}_i \boldsymbol{K}^\top)_j.$$

To rank keys globally, these per-query attention weights must be aggregated across queries into a single importance score per key, after which the top-t keys are retained.

We consider three aggregation functions:

$$s_j^{\text{mean}} = \frac{1}{n} \sum_{i=1}^{n} a_{i,j}, \quad s_j^{\text{rms}} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} a_{i,j}^2}, \quad s_j^{\text{max}} = \max_i a_{i,j}.$$

Figure 9 compares downstream performance across these choices for four baseline methods as well as our Highest Attention Keys approach. There is no choice that is consistently best across methods, though RMS offers a good balance and we adopt it in our method.

F.2. Reconstruction vs. Downstream Accuracy

In addition to QA accuracy, we also compute loss (log-perplexity) on the *original-cache generations*: we evaluate the negative log-likelihood of responses sampled from the full cache, but conditioned on the compacted cache. This metric measures how well a compacted cache preserves the model's token-level behavior, differing from the forward KL divergence only by a constant additive term. Figure 1 (left) replicates our main trade-off plot using this loss metric, and Figure 10 (right) compares downstream multiple-choice accuracy directly against log-perplexity across methods and compaction ratios.

Across attention-matching methods and token-selection baselines, we observe that points roughly interpolate between the *no-context* results (high loss, low accuracy) and the *full-context* results (low loss, high accuracy). As expected, improvements in reconstruction tend to translate into improvements in QA accuracy.